

The Squeeze Theorem:

Let f , g and h be functions satisfying

$$g(x) \leq f(x) \leq h(x)$$

for all x in some open interval containing the number a ,
~~with~~ except possibly at a itself. ~~Suppose that for example~~
If g and h have the same limit as x approaches a , say

$$\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x) = L$$

then f also has this limit as x approaches a , i.e.

$$\lim_{x \rightarrow a} f(x) = L.$$

Now our previous problem.

Problem: Estimate $\lim_{x \rightarrow 0} x \sin(\frac{1}{x})$ if exists.

Solution: Assume that $f(x) = x \sin(\frac{1}{x})$

Since $-1 \leq \sin(\frac{1}{x}) \leq 1$

$$-|x| \leq |x| \sin(\frac{1}{x}) \leq |x|$$

$$\lim_{x \rightarrow 0} -|x| \leq \lim_{x \rightarrow 0} |x| \sin(\frac{1}{x}) \leq \lim_{x \rightarrow 0} |x|$$

again, $|x| \rightarrow 0$ as $x \rightarrow 0$ i.e. $\lim_{x \rightarrow 0} |x| = 0$

Applying the squeeze theorem,

$$\lim_{x \rightarrow 0} x \sin(\frac{1}{x}) = 0$$

✱

Why we don't use the
properties of limit?

$$\lim_{x \rightarrow 0} x \cdot \lim_{x \rightarrow 0} \sin(\frac{1}{x}) \stackrel{?}{=} 0$$

explain!!!

Some Extra Problems

A [Using Squeeze Th^m]

1. Compute: $\lim_{x \rightarrow \infty} \frac{2 - \cos x}{x+3}$

Ans: 0

2. Compute $\lim_{x \rightarrow \infty} \frac{x^2(2 + \sin^2 x)}{x+100}$

Ans: ∞ (does not exist)

3. Compute $\lim_{x \rightarrow 0^-} x^3 \cos\left(\frac{2}{x}\right)$

Ans: 0

4. Compute $\lim_{x \rightarrow -\infty} \frac{5x^2 - \sin(3x)}{x^2 + 10}$

Ans: 5

13. Find the limit if it exist, otherwise explain why it does not exist:

(a) $\lim_{x \rightarrow -3} \frac{|x^2 - 9|(x+2)}{x^2 + 7x + 12}$

(b) $\lim_{x \rightarrow 4} \frac{16 - x^2}{x^3 - 5x^2 + 4x}$

(c) $\lim_{x \rightarrow 0} \frac{x}{3 - \sqrt{x+9}}$

(d) $\lim_{x \rightarrow 0} f(x)$ where $f(x) = \begin{cases} e^{-|x|/2} & , -1 < x < 0 \\ x^2 & , 0 < x < 2 \end{cases}$

(e) $\lim_{t \rightarrow -1} \frac{t+1}{|t+1|}$

(f) $\lim_{x \rightarrow -3} \frac{\sqrt{2x+22} - 4}{x+3}$